ADVANCED MACHINE LEARNING · WORKSHEET 01

The ML Project Lifecycle (Part 1)

Problem Definition · Data Collection · Exploratory Analysis · Preprocessing

Lecture 1

Name:	**Branch / Year:**	**Date:**

THE ML PROJECT LIFECYCLE

PART 1 DEFINING THE PROBLEM (THE FOUNDATION)

▷ **HOOK**

Your music streaming company says: "Users are leaving! Build an AI to fix this!" If you open a laptop and start coding a linear regression right now, why are you almost guaranteed to fail?

The five-step blueprint

An algorithm cannot guess what your business needs. Phase 1 translates a messy human problem into an **explicit mathematical framework** using five steps:

Step	**What You Do**	**Example**
1. Understand the Problem	Isolate the exact friction point	Users canceling premium subscriptions = churn
2. Set Quantifiable Goals	Define a measurable target	"Reduce churn by 10% over two quarters"
3. Assess ML Feasibility	Ask: do we need ML? Do we have data?	Can a SQL query solve it? If yes, skip ML.
4. Identify Constraints	Budget, latency, privacy laws (GDPR)	Predictions in <10 ms on a phone?
5. Stakeholder Alignment	PMs, devs, execs agree on success	Shared definition of "success"

Blueprint in action

- **Vague:** "We want to improve our online retail store."
- **Rigorous:** "Using historical clickstreams and purchase data, predict which users will stop shopping within 3 months."

★ KEY INSIGHT

A poorly defined problem wastes more time than a bad algorithm. You cannot optimize "make it better" the algorithm needs a number to minimize.

▷ PRACTICE P1 · Rewrite each vague request as a rigorous ML problem statement

(a) "Our hospital needs better patient care."

Expected answer: "Using patient vitals, lab results, and admission records from the past 5 years, predict which ICU patients are at risk of readmission within 30 days, targeting a 15% reduction in preventable readmissions."

(b) "Make our email system smarter."

Expected answer: "Using email metadata (sender, subject, timestamps) and user-labelled spam/non-spam data, classify incoming emails as spam or not-spam with $\geq$98% precision and $\geq$95% recall."

△ REFLECT 1

A company has excellent customer data, but it is not permitted to use the data for training an ML model. At which phase of the Machine Learning Blueprint does the project encounter this problem?

Hint: Operational Constraints.

Expected answer: Phase 4: Identify Constraints. Even though data exists, legal/regulatory constraints (e.g., GDPR, internal data governance policies) prevent its use for ML training. This must be identified before investing effort in model building.

⧫ TAKEAWAY

Never touch code until you have a quantified target, confirmed data access, and stakeholder agreement.

PART 2 DATA COLLECTION (SOURCING THE FUEL)

▷ HOOK

Can you build an ML model without any data? An algorithm has no concept of reality it only knows the world you present via training data. Flawed collection = broken model.

Five production data sources

Source	Example
Internal Warehouses (SQL databases)	Customer purchase history, transactional logs
Public Repositories	Kaggle, UCI ML Repository
APIs	Live stock prices, weather feeds
Web Scraping	Product prices from competitor sites
Manual Labeling	Radiologists labeling tumors on X-rays

Data structures

Type	Structure	Example
Structured	Rigid rows & columns (SQL)	Order ID, Timestamp, Amount, User ID
Unstructured	No grid format	Images, audio, video, free-text reviews
Semi-structured	Tags/keys but no strict schema	JSON, XML, IoT sensor logs
Time-series	Indexed over time intervals	Daily stock prices, hourly weather data

Feature types

Type	Nature	Example
Categorical (Nominal)	Labels, no order	Color, City, Blood Type (A/B/AB/O)
Categorical (Ordinal)	Labels with rank	Education: HS $<$ Bachelor $<$ Master $<$ PhD
Numerical (Discrete)	Countable integers	Number of rooms, items in cart
Numerical (Continuous)	Infinitely divisible	Income, temperature, weight

★ KEY INSIGHT

An algorithm only knows the world you show it. Garbage In = Garbage Out.

▷ PRACTICE P2 · Classify each feature: Structured / Unstructured / Semi / Time-Series and Categorical / Numerical

(a) A food delivery app stores customer ratings as 1, 2, 3, 4, or 5 stars.
Structure: **Structured** Feature: **Categorical (Ordinal)**

(b) A hospital records patient body temperature every hour for 7 days after surgery.
Structure: **Time-series** Feature: **Numerical (Continuous)**

(c) An e-commerce platform stores each product's description written by the seller in free text.
Structure: **Unstructured** Feature: **N/A (text)**

(d) A government database stores the PIN code (postal code) of each citizen's address.
Structure: **Structured** Feature: **Categorical (Nominal)**

(e) A weather station logs wind speed in km/h every 10 minutes for an entire year.
Structure: **Time-series** Feature: **Numerical (Continuous)**

△ REFLECT 2

Your scraper collects 100k reviews but 40% are bot-generated spam. Is more data always better?

Hint: quality vs. quantity.

Expected answer: No. More data is not always better if it contains noise, bias, or irrelevant samples. Here, 40% bot-generated spam would degrade model quality. Data quality (clean, representative, relevant) matters more than raw volume. You should filter or remove the spam before training.

♦ TAKEAWAY

Know your sources, know the structure, know the feature types before you write a single line of code.

PART 3 EXPLORATORY DATA ANALYSIS (EDA)

▷ HOOK

Your company hands you 500,000 customer records. What is the very first thing you do? If you said "train a model" stop. A doctor does not prescribe treatment before examining the patient.

EDA is the **diagnostic stage**. It transforms raw data into actionable insights before any preprocessing.

EDA checklist

Goal	**What You Check**
1. Dataset Structure	Rows, columns, data types, feature names
2. Missing Values	Which columns? How many? Random or systematic?
3. Outliers	Extreme values (data error? fraud? rare case?)
4. Duplicates	Repeated rows from collection/merging errors
5. Distributions	Normal (bell) vs. Skewed (left/right)
6. Relationships	Correlation, dependency, redundant features
7. Class Imbalance	95% Non-Fraud vs. 5% Fraud → accuracy lies

Analysis levels

- **Univariate:** One variable (distribution, mean, median, IQR, outliers). Tool: Histogram, Box Plot.
- **Bivariate:** Two variables (correlation, trends). Tool: Scatter Plot, Correlation Heatmap.
- **Multivariate:** Many variables (interaction effects). Tool: Heatmap, Pair Plot.

Common visualizations

Chart	**Use For**
Histogram	Distributions, skewness
Box Plot	Outliers, spread (Q_1, Q_3, IQR)
Bar Chart	Categorical comparisons
Scatter Plot	Relationship between two numerical variables
Correlation Heatmap	Feature correlations ($+1$, 0, -1)

Correlation

- **Positive:** both increase together (Experience ↔ Salary)
- **Negative:** one up, other down (Price ↔ Demand)
- **None:** no clear relationship

★ KEY INSIGHT

Skip EDA and you risk training on missing values, outliers, duplicates, and imbalanced classes all of which silently destroy performance. EDA is not optional.

▷ PRACTICE P3 · Given this dataset, list your EDA steps and pick the right chart

Dataset: 100,000 rows columns: Customer_ID, Age, Gender, Annual_Income, Spending_Score, City.

First 3 EDA steps you perform:

Expected answer: 1. Check dataset shape, data types, and summary statistics (`df.info()`, `df.describe()`).
2. Check for missing values and duplicates (`df.isnull().sum()`, `df.duplicated().sum()`).
3. Visualize distributions and check for outliers (histograms for Age/Income, box plots for Spending_Score).

Chart for Age distribution? **Histogram**
Chart for Income vs. Spending_Score? **Scatter Plot**
Chart to check if Gender affects Spending_Score? **Bar Chart / Box Plot**

△ REFLECT 3

A dataset is 95% Class A and 5% Class B. A model predicts A every time, scoring 95% accuracy. Is this useful?

Hint: what is the recall for Class B?

Expected answer: No. The model has 0% recall for Class B it never detects the minority class. Accuracy is misleading when classes are imbalanced. A useful model must also correctly identify Class B instances. Metrics like precision, recall, F1-score, or AUC-ROC are more informative in such scenarios.

♦ TAKEAWAY

EDA first, algorithm later. Understand structure → find issues → visualize patterns → then preprocess.

PART 4 DATA PREPROCESSING (THE REAL ENGINEERING WORK)

▷ HOOK

In textbooks, data is always clean. In the real world? Messy, biased, incomplete. Can a sophisticated algorithm self-correct and ignore the garbage? No. Garbage In, Garbage Out.

Algorithms see data as matrices of numbers. Missing value? Equation breaks. One feature in millions, another in single digits? The large one overwhelms the math. Preprocessing builds a **clean numerical matrix**.

Step 1: Handling missing values

Strategy	Action	When to Use	Danger
Deletion	Drop rows or columns	<1–2% missing; or column >80% empty	Discards info, introduces bias
Imputation	Fill with mean/median (num) or mode (cat)	Moderate missingness	May introduce artificial patterns

Step 2: Handling inconsistent data

Type	Issue	Fix
Mixed Units	Weight in both kg and lbs	Standardize to single unit
Typos	'Californa', 'California', 'CA'	Regex / Levenshtein distance matching
Duplicates	Same record logged twice	Remove duplicate records
Inconsistent Labels	Yes, Y, True, 1 all mean "yes"	Map all to single label format

Step 3: Feature scaling

When one feature ranges from 1–5 (bedrooms) and another \$20,000–\$500,000 (income), distance-based algorithms (KNN, SVM) and gradient descent are dominated by the larger feature. Scaling fixes this.

Technique	Formula	Use When	Risk
A. Min-Max (Normalization)	$X_{\text{scaled}} = \dfrac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$ Maps to $[0, 1]$	Not Gaussian; neural nets; bounded ranges	One outlier squashes everything near 0
B. Z-Score (Standardization)	$Z = \dfrac{X - \mu}{\sigma}$ Mean→0, Std→1	Data follows Normal distribution	Performs best when features are approximately Gaussian.
C. Max Absolute	$X_{\text{scaled}} = \dfrac{X}{\max(\lvert X \rvert)}$ Maps to $[-1, 1]$	Sparse data (many zeros, e.g., bag-of-words)	Preserves zeros and sign
D. Robust	$X_{\text{scaled}} = \dfrac{X - \text{Median}}{Q_3 - Q_1}$ $\text{IQR} = Q_3 - Q_1$	Heavy outliers you cannot drop	Ignores mean/std; uses median + middle 50%

★ KEY INSIGHT

Min-Max is sensitive to outliers (one extreme value squashes everything). Z-Score assumes normality. Robust Scaling is your safety net when outliers are unavoidable.

▷ PRACTICE P4 · Compute by hand

A. MIN-MAX SCALING

Dataset $X = [10, 20, 30, 40, 50]$. $X_{\text{min}} = 10$, $X_{\text{max}} = 50$.

Scale $X = 10$: $(10 - 10)/(50 - 10) = 0.0$ Scale $X = 30$: $(30 - 10)/(50 - 10) = 0.5$ Scale $X = 50$: $(50 - 10)/(50 - 10) = 1.0$

B. Z-SCORE STANDARDIZATION

Dataset $X = [2, 4, 6, 8, 10]$. $\mu = 6$, $\sigma = 2\sqrt{2} \approx 2.83$.

z for $X = 2$: $(2-6)/2.83 \approx -1.41$ z for $X = 6$: $(6-6)/2.83 = 0$ z for $X = 10$: $(10-6)/2.83 \approx 1.41$

C. ROBUST SCALING (full walkthrough) *(Optional)*
Dataset $X = [100,\ 150,\ 200,\ 250,\ 50000]$.

Step 1 Sort: [**100** , **150** , **200** , **250** , **50000**]
Step 2 Median (Q_2): **200**
Step 3 Q_1 (median of lower half $[100, 150]$): **125**
Step 4 Q_3 (median of upper half $[250, 50000]$): **25125**
Step 5 IQR $= Q_3 - Q_1 =$ **25125** $-$ **125** $=$ **25000**
Step 6 Scale $X = 100$: (100− **200**) / **25000** $=$ -0.004
Step 7 Scale $X = 50000$: (50000− **200**) / **25000** $=$ 1.992

Expected answer: The outlier (50000) didn't break the scale because Robust Scaling uses Median and IQR instead of Mean and Std. The outlier affects Q_3 but the IQR-based denominator is far more stable than $(X_{\max} - X_{\min})$, so normal values retain meaningful variation.

▷ PRACTICE P5 · Pick the scaler and justify in one line

(a) Salary data: \$20k–\$5M with CEO outliers at \$50M.
Scaler: **Robust** Why: **Heavy outliers; Robust uses median/IQR**

(b) Pixel values for images, range $[0, 255]$, no outliers.
Scaler: **Min-Max** Why: **Bounded range, no outliers, maps to** $[0, 1]$

△ REFLECT 4

Why does Min-Max scaling fail when one value is 50,000 and the rest are under 300?

Hint: what happens to $(X_{\max} - X_{\min})$ *in the denominator?*

Expected answer: The denominator $(X_{\max} - X_{\min})$ becomes very large (≈49,700). This squashes all the normal values (under 300) into a tiny range near 0, while only the single outlier maps to 1. The scaled data loses all meaningful variation among the majority of points. Robust Scaling (using median and IQR) is preferred when outliers are present.

⧫ TAKEAWAY

Match the scaler to the data: Min-Max for bounded/clean, Z-Score for Gaussian, Robust for outliers, Max Abs for sparse.

Step 4: Categorical encoding

If we map Red=0, Green=1, Blue=2, the model thinks Blue > Green > Red, and that (Red+Blue)/2 = Green. That is a **mathematical lie**. Choose encoding carefully.

Encoding	How	Safe For	Danger
Label	Unique integer per category Red=0, Blue=1, Green=2	Binary targets (Spam=1/0); tree-based models (split, not distance)	Implies false order for distance-based models (KNN, SVM)
Ordinal	Controlled rank mapping Low=0, Med=1, High=2	Categories with natural rank; tree + linear models	Wrong if categories are actually unordered
One-Hot	N binary columns (1=present) Red→[0,0,1], Blue→[1,0,0]	Unordered nominal data with few unique values	Curse of Dimensionality with many categories
Binary	Integer → binary digits → split across $\log_2 N$ columns	Many categories; space-efficient alternative	Lower dimensional representation than One-Hot

One-Hot worked:

Input	Color_Blue	Color_Green	Color_Red
Red	0	0	1
Blue	1	0	0
Green	0	1	0

Binary worked (Blue=1→01, Red=2→10, Green=3→11):

Input	Bit_0	Bit_1
Red (10)	1	0
Green (11)	1	1
Blue (01)	0	1

★ KEY INSIGHT

One-Hot: 3 categories = 3 columns. Binary: 3 categories = 2 columns. For 1000 categories, One-Hot = 1000 cols; Binary ≈ 10 cols.

▷ PRACTICE P6 · Pick the encoding and justify

(a) 500 city names, algorithm = Logistic Regression.
Encoding: **Binary** Why: **500 categories; One-Hot = 500 cols; Binary ≈ 9 cols**

(b) Education (HS < Bachelor < Master < PhD), algorithm = Linear Regression.
Encoding: **Ordinal** Why: **Natural rank exists; preserves order for linear model**

(c) Color (Red, Blue, Green), algorithm = KNN.
Encoding: **One-Hot** Why: **Nominal, few values, KNN uses distances**

▷ PRACTICE P7 · Binary encoding by hand

Categories: [Cat, Dog, Fish, Bird]. Assign integers 1–4, convert to binary, fill the table.

Cat = 1 → binary **01** Dog = 2 → binary **10**
Fish = 3 → binary **11** Bird = 4 → binary **100**

How many bits needed? **2 (or 3)** How many columns would One-Hot need? **4**

△ REFLECT 5

Why can Label Encoding mislead KNN but not a Decision Tree?

Hint: think about distances vs. splits don't worry, this will be covered later.

Expected answer: KNN uses distance calculations (e.g., Euclidean distance) between data points. Label Encoding assigns arbitrary integers (Red=0, Blue=1, Green=2), creating false numerical relationships KNN thinks Blue is "closer" to Red than Green is. Decision Trees only use threshold-based splits (e.g., $x \leq 1$), so the actual magnitude and ordering of encoded values does not affect the split logic. Trees treat each split independently without computing distances.

⧫ TAKEAWAY

Ordinal for ranked categories, One-Hot for nominal with few values, Binary for nominal with many. Label only for tree models or binary targets.

SUMMARY OF THE LECTURE

#	Phase	Key Idea
1	Define the Problem	Vague goal → rigorous ML objective
2	Collect Data	Internal DBs, APIs, scraping, labeling
3	Exploratory Data Analysis	Structure, missing, outliers, distributions
4	Data Preprocessing	Clean, scale, encode → numerical matrix

BY THE END OF THIS SHEET YOU CAN

- ☐ Translate a vague business request into a constrained ML problem statement with quantifiable goals.
- ☐ Classify data sources and distinguish structured, unstructured, semi-structured, and time-series data.
- ☐ Perform EDA: detect missing values, outliers, duplicates, class imbalance, and interpret correlations.
- ☐ Select and compute the correct feature scaling (Min-Max, Z-Score, Robust) by hand.
- ☐ Choose the right categorical encoding (Label, Ordinal, One-Hot, Binary) and justify your choice.